Chapter 1. Time Series Data

Dependence, stationarity and autocovariance

Contents

1.1 Time series data with dependence **2**

1.1.1 Trend and moving-average smoothing 2

1.1.2 Exponential smoothing 2

1.1.3 Trend, seasonality and noise 3

1.2 Stationary time series **3**

1.1 Time series data with dependence

Motivation. Introductory statistics rests on the i.i.d. assumption, but observations recorded over time (temperatures, stock prices, economic indicators) are typically correlated with their own past. Time series analysis studies such dependent data; this section shows the simplest descriptive devices (smoothing) and explains why a model for the dependence is needed.

1.1.1 Trend and moving-average smoothing

Let $x_1, \dots, x_n$ be observations, e.g. monthly closing values of a stock index. Raw prices fluctuate so much that the underlying movement is hard to see. The *moving average* of span m,

$$y_t = \frac{x_{t-m+1} + \cdots + x_t}{m},$$

replaces each value by the mean of the last m observations. Larger m gives a smoother curve; a short-span average describes short-run movement, a long-span one the long-run trend. Producing such a curve is called *smoothing*.

To see why it works, suppose the data follow the signal-plus-noise model

$$x_t = f(t) + z_t, \qquad t = 1, \dots, n, \tag{1.1}$$

where f is an unknown deterministic function and z_t are i.i.d. errors with mean 0 and variance σ^2. Then

$$y_t = \frac{f(t-m+1) + \cdots + f(t)}{m} + \frac{z_{t-m+1} + \cdots + z_t}{m}. \tag{1.2}$$

For moderately large m the second term is close to $Ez_1 = 0$ by the law of large numbers (its variance is σ^2/m), so $y_t \approx f_m(t) := m^{-1} \sum_{j=0}^{m-1} f(t-j)$, a local average of the trend.

1.1.2 Exponential smoothing

An alternative to equal weights is to discount the past geometrically. For $0 < w < 1$ put

$$S_t = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j}.$$

Given an initial value S_0^*, define recursively

$$S_t^* = w x_t + (1-w) S_{t-1}^*. \tag{1.3}$$

Iterating (1.3) gives

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^* = S_t + (1-w)^t S_0^*, \tag{1.4}$$

so $S_t^* \approx S_t$ for large t, and the cheap recursion may be used instead of S_t. Common choices of S_0^* are x_1 or the sample mean; w can be chosen to minimize the one-step prediction error

$$SSE(w) = \sum_{t=2}^{n} (x_t - S_{t-1}^*)^2. \tag{1.5}$$

1.1.3 Trend, seasonality and noise

Writing T_t for the trend $f(t)$, one also uses the decomposition $x_t = T_t + S_t + z_t$, where the *seasonal component* satisfies $S_t = S_{t+s}$ for a period s ($s = 4$ for quarterly, $s = 12$ for monthly data). The i.i.d. assumption on z_t is only a convenience: real errors are usually strongly correlated. The next section introduces stationarity, the basic framework for modelling such dependence.

1.2 Stationary time series

Motivation. To do inference from a single realization we need the probabilistic structure to be stable over time. Stationarity formalizes this, and the autocovariance function becomes the central object describing the dependence.

Definition 1.2.1 (Time series)

If T is the set of integers $\mathbb{Z}$ or of positive integers $\mathbb{Z}^+ = \{1, 2, \dots\}$, a collection of random variables $\{X_t : t \in T\}$ is called a *time series*.

Definition 1.2.2 (Strictly stationary time series)

$\{X_t\}$ is *strictly stationary* if for all $k \geq 1$, all integers $t_1, \dots, t_k$ and h, and all real $x_1, \dots, x_k$,

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k).$$

That is, every finite-dimensional distribution is invariant under time shifts.

Definition 1.2.3 (Weakly stationary time series)

$\{X_t\}$ is *(weakly) stationary* if
(1) $EX_t = \mu$ for all t;
(2) $EX_t^2 < \infty$ for all t;
(3) $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_0, X_h)$ for all t, h.

Condition (1) says the level is constant, (2) that variances exist, and (3), the essential one, that the covariance between two observations depends only on their time distance; in particular the variance is constant. Unless stated otherwise, "stationary" means weakly stationary.

Example 1.2.4 (Strict versus weak stationarity)

(1) An i.i.d. sequence $\{X_t\}$ is strictly stationary.
(2) If moreover $EX_1^2 < \infty$, it is also weakly stationary. An i.i.d. Cauchy sequence is strictly but not weakly stationary, since the variance does not exist.
(3) If ϵ_t are i.i.d. with mean 0, variance 1, then $X_t = \epsilon_t + \epsilon_{t-1}$ is strictly and weakly stationary.

Neither notion contains the other, but a strictly stationary series with finite variance is weakly stationary: shift invariance of the bivariate law of (X_t, X_{t+h}) gives (1) and (3). Throughout, stationary series are assumed to have finite variance.

Definition 1.2.5 (Autocovariance function)

For a stationary $\{X_t\}$ the real function

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \qquad h = 0, \pm 1, \pm 2, \ldots,$$

is called the *autocovariance function* (ACVF).

Theorem 1.2.6 (Basic properties of the ACVF)

The autocovariance function satisfies
(1) $\gamma(0) \geq 0$;
(2) $\gamma(-h) = \gamma(h)$;
(3) $|\gamma(h)| \leq \gamma(0)$.

Proof. [Supplement]

Step 1. $\gamma(0) = \mathrm{Var}(X_t) \geq 0$.

Step 2. By stationarity and symmetry of covariance, $\gamma(-h) = \mathrm{Cov}(X_t, X_{t-h}) = \mathrm{Cov}(X_{t-h}, X_t) = \gamma(h)$, the last equality by putting $s = t - h$.

Step 3. By the Cauchy–Schwarz inequality, $|\mathrm{Cov}(X_t, X_{t+h})| \leq \sqrt{\mathrm{Var}\, X_t \ \mathrm{Var}\, X_{t+h}} = \gamma(0)$. ■

Theorem 1.2.7 (Nonnegative definiteness)

For any reals $x_1, \ldots, x_k$ and integers $t_1, \ldots, t_k$,

$$\sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j \geq 0.$$

Proof. [Supplement] Let $Y = \sum_{i=1}^{k} x_i X_{t_i}$. Since $\mathrm{Cov}(X_{t_i}, X_{t_j}) = \gamma(t_j - t_i) = \gamma(t_i - t_j)$ by Theorem 1.2.6(2),

$$0 \leq \mathrm{Var}(Y) = \sum_{i,j=1}^{k} x_i x_j \,\mathrm{Cov}(X_{t_i}, X_{t_j}) = \sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j.$$

■

In matrix form: the covariance matrix $\Gamma_k = [\gamma(t_i - t_j)]_{i,j=1}^{k}$ of any finite sub-vector is nonnegative definite. (Conversely, every even nonnegative definite function on $\mathbb{Z}$ is the ACVF of some stationary, indeed Gaussian, series.)

Definition 1.2.8 (White noise)

A stationary series $\{X_t\}$ whose ACVF satisfies $\gamma(h) = 0$ for $h \neq 0$ is called *white noise*, written $\{X_t\} \sim WN(0, \sigma^2)$ with $\sigma^2 = \gamma(0)$ (mean zero is understood). If in addition the X_t are i.i.d., we write $\{X_t\} \sim IID(0, \sigma^2)$.

Definition 1.2.9 (Autocorrelation function)

For a stationary $\{X_t\}$,

$$\rho(h) = \gamma(h)/\gamma(0)$$

is the *autocorrelation function* (ACF); for fixed h, $\rho(h)$ is the autocorrelation coefficient at lag h.

By Theorem 1.2.6, $\rho(0) = 1$, $\rho(-h) = \rho(h)$ and $|\rho(h)| \le 1$.

Definition 1.2.10 (Multivariate normal distribution)

Let $\mu = (\mu_1, \dots, \mu_k)^T \in \mathbb{R}^k$ and let $\Sigma = (\sigma_{ij})_{i,j=1}^k$ be symmetric positive definite: $\sigma_{ij} = \sigma_{ji}$ and $a^T \Sigma a > 0$ for all $a \in \mathbb{R}^k \setminus \{0\}$. If the random vector $Z = (Z_1, \dots, Z_k)^T$ has joint density

$$f(z) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\Big\{-\tfrac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\Big\}, \qquad z \in \mathbb{R}^k,$$

then Z follows the *multivariate normal distribution* with mean μ and covariance Σ; here $\mu_i = EZ_i$ and $\sigma_{ij} = \mathrm{Cov}(Z_i, Z_j)$.

Definition 1.2.11 (Gaussian time series)

$\{X_t\}$ is a *Gaussian time series* if $(X_{t_1}, \dots, X_{t_k})^T$ is multivariate normal for every k and all $t_1, \dots, t_k$.

Since a normal law is determined by its mean and covariance, a weakly stationary Gaussian series is strictly stationary. Normality rarely holds exactly, but it makes the theory tractable and is the natural starting point for extensions. A key consequence concerns prediction: for a zero-mean stationary Gaussian series,

$$E(X_{k+1} \mid X_1, \dots, X_k) = \sum_{i=1}^{k} a_i X_i$$

for some reals a_i, i.e. the best predictor is linear. Given data $X_1, \dots, X_n$, such coefficients can be estimated as in regression, by minimizing $\sum_{t=k+1}^{n} (X_t - a_1 X_{t-1} - \cdots - a_k X_{t-k})^2$.

[Supplement] Proof of linearity of the conditional mean. Let $\mathbf{X} = (X_1, \dots, X_k)^T$, $\Gamma_k = \mathrm{Cov}(\mathbf{X})$ (assume invertible) and $\gamma_k = \mathrm{Cov}(\mathbf{X}, X_{k+1})$. Put $a = \Gamma_k^{-1}\gamma_k$ and $U = X_{k+1} - a^T\mathbf{X}$. Then $\mathrm{Cov}(\mathbf{X}, U) = \gamma_k - \Gamma_k a = 0$. Since $(\mathbf{X}, U)$ is jointly normal (a linear image of a normal vector), zero covariance implies U is independent of $\mathbf{X}$, so $E(U \mid \mathbf{X}) = EU = 0$. Hence $E(X_{k+1} \mid \mathbf{X}) = a^T\mathbf{X} + E(U \mid \mathbf{X}) = a^T\mathbf{X}$. ■

Stationarity is the backbone of time series analysis. Real data are often nonstationary, but transformations (e.g. differencing, leading to the ARIMA models of Chapter 5) can frequently reduce them to stationary ones.